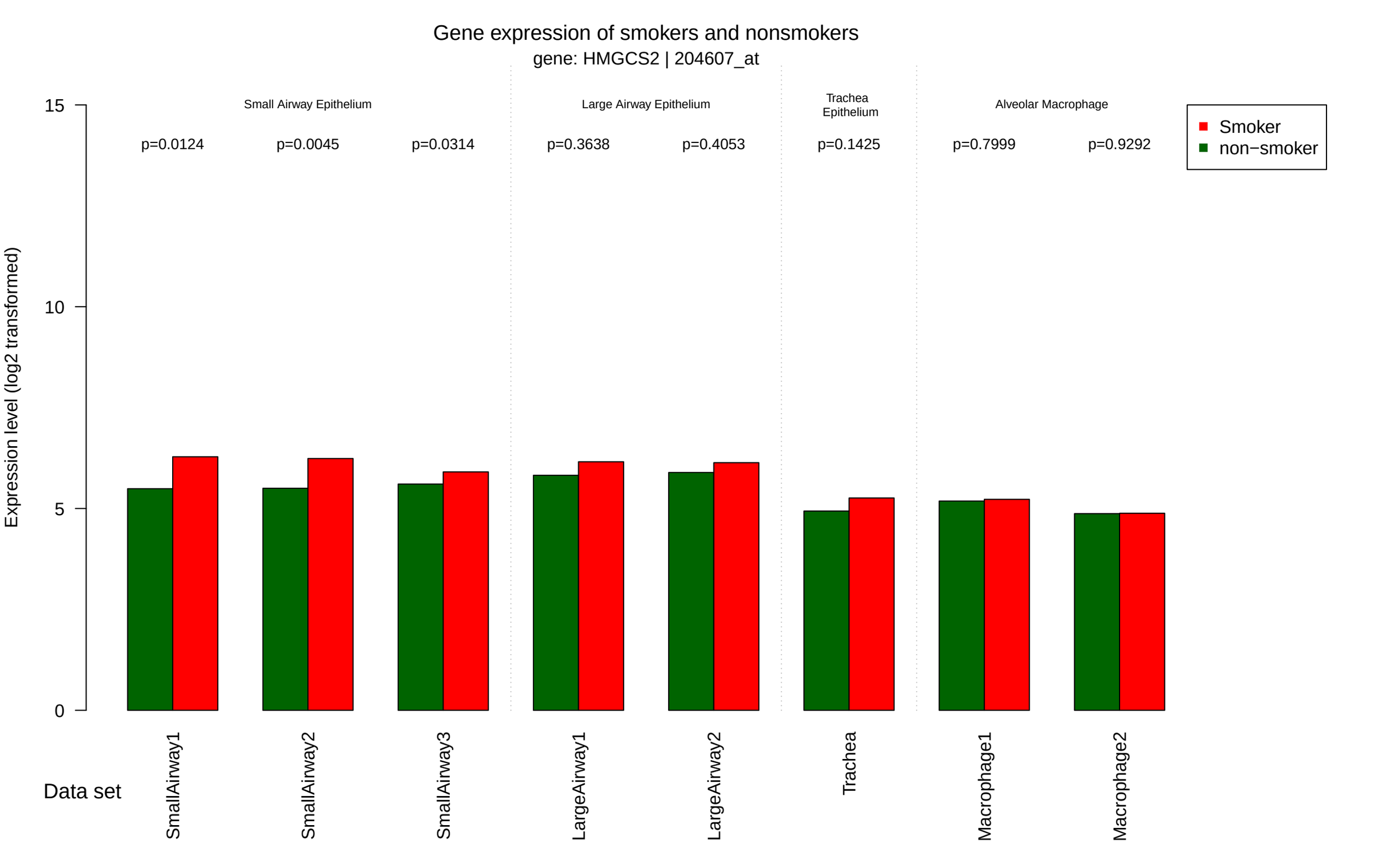
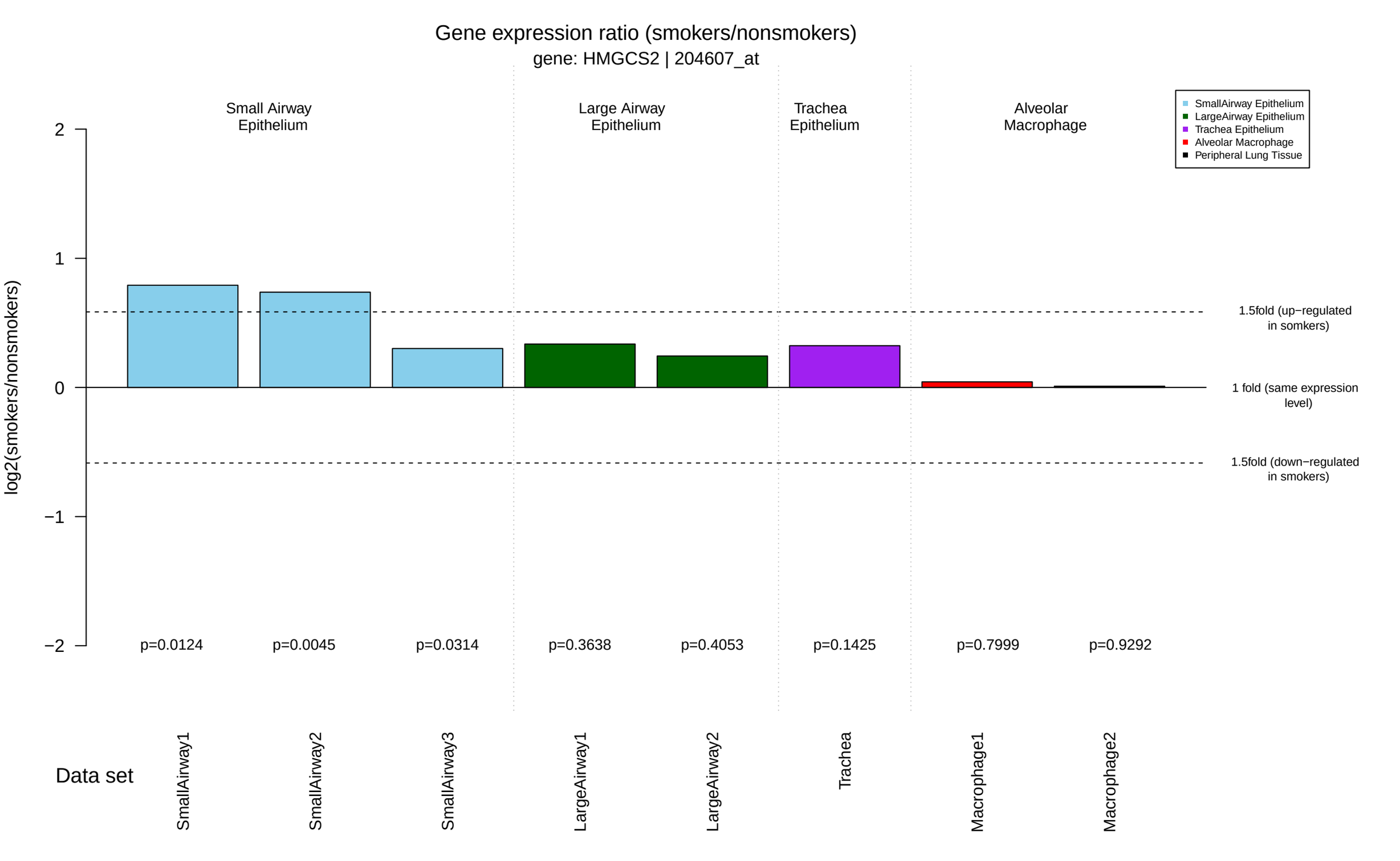
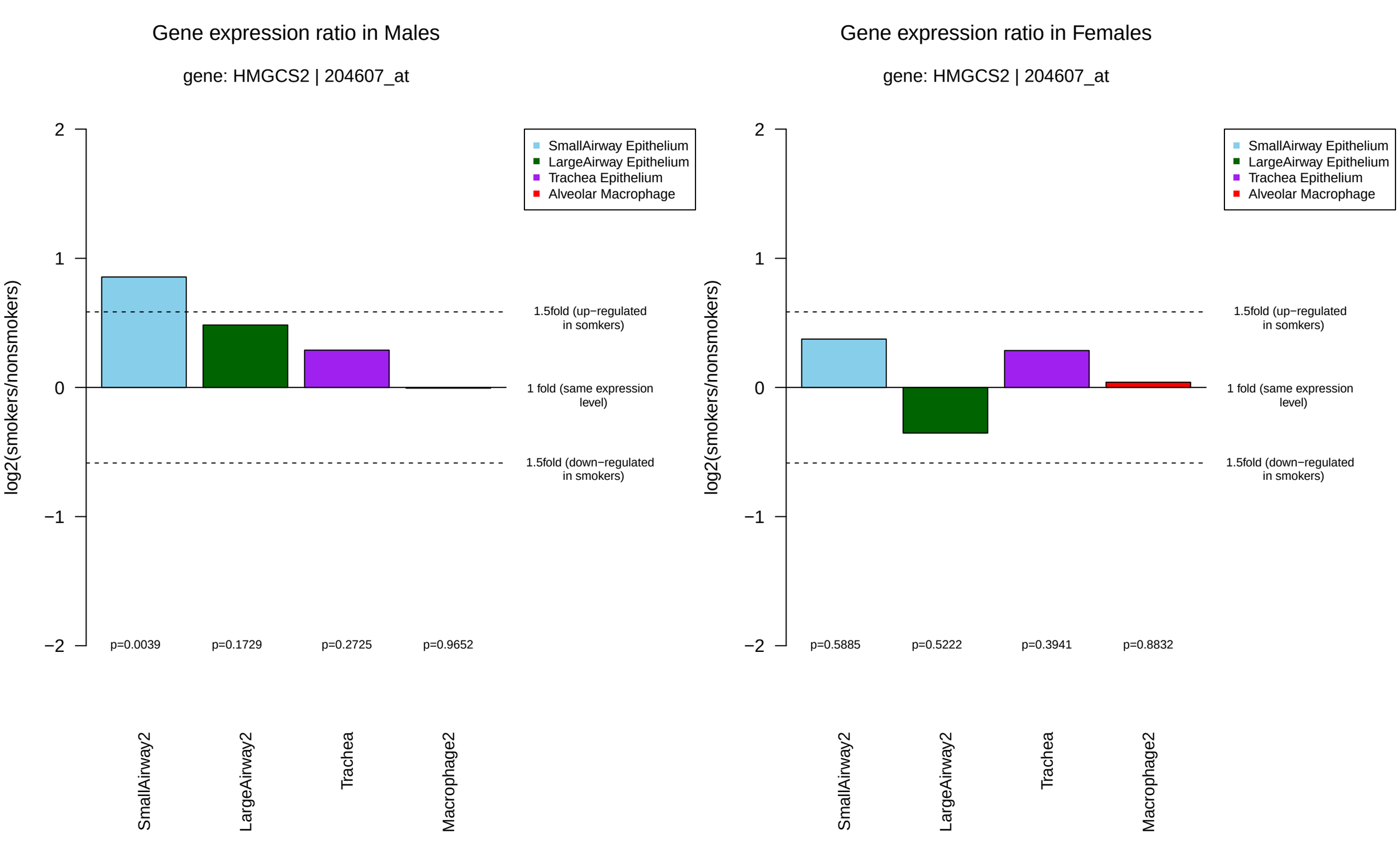
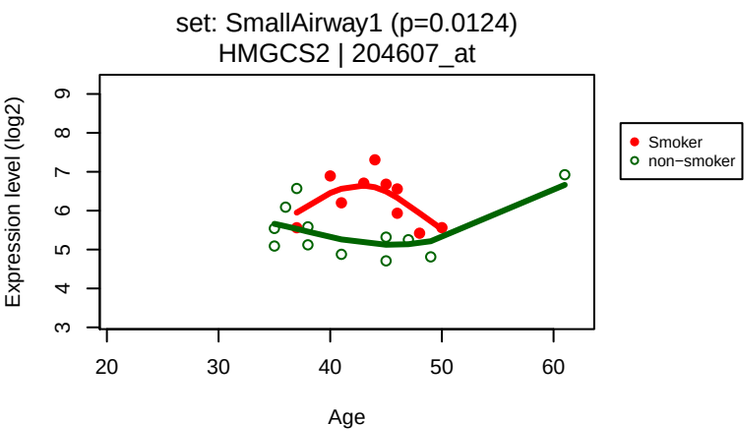
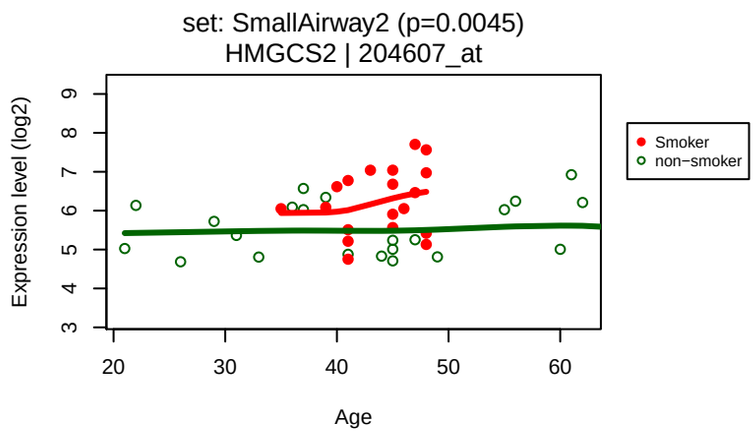
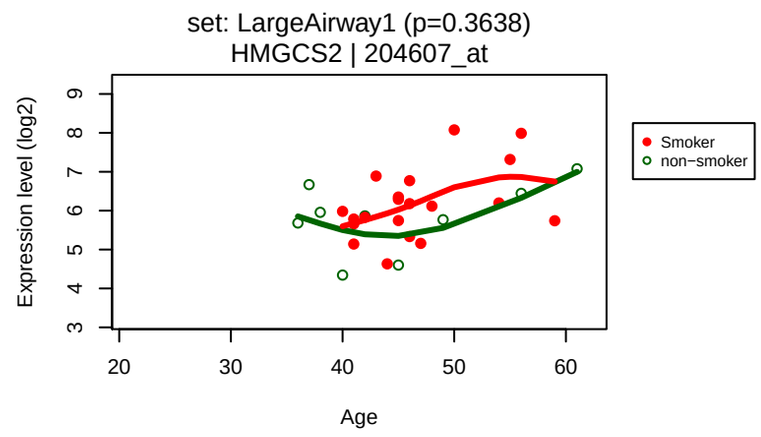

Gene expression of smokers and nonsmokers
gene: HMGCS2 | 204607_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0124
p=0.0045
p=0.0314
p=0.3638
p=0.4053
p=0.1425
p=0.7999
p=0.9292
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

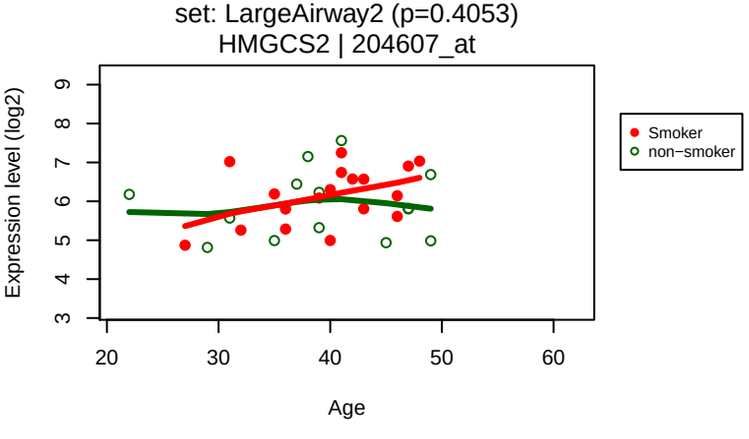

Gene expression ratio (smokers/nonsmokers)
gene: HMGCS2 | 204607_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0124
p=0.0045
p=0.0314
p=0.3638
p=0.4053
p=0.1425
p=0.7999
p=0.9292
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

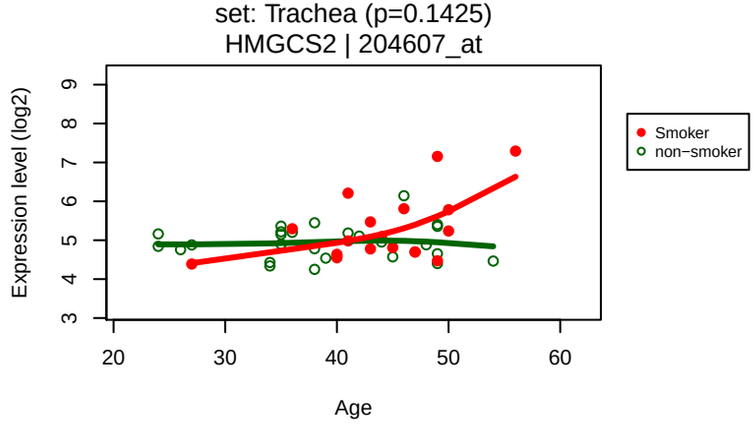

Gene expression ratio in Males
gene: HMGCS2 | 204607_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0039
p=0.1729
p=0.2725
p=0.9652
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HMGCS2 | 204607_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5885
p=0.5222
p=0.3941
p=0.8832
SmallAirway2
LargeAirway2
Trachea
Macrophage2

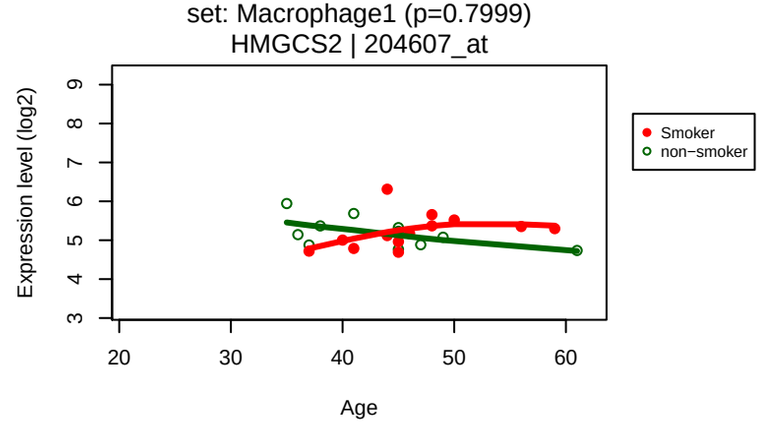

set: SmallAirway1 (p=0.0124)
HMGCS2 | 204607_at
Expression level (log2)
Age
Smoker
non−smoker

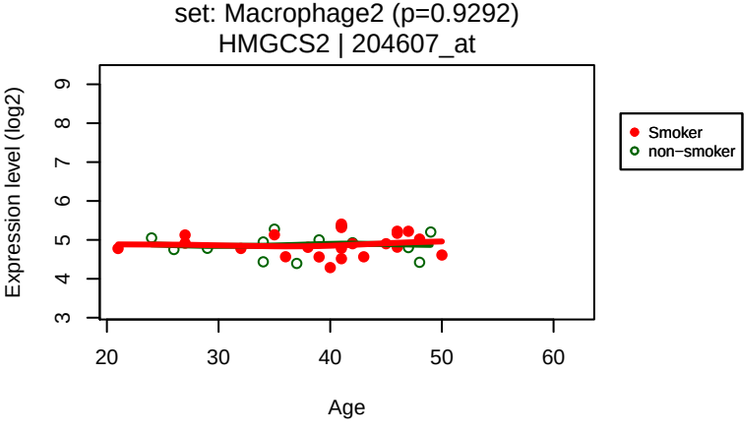

set: SmallAirway2 (p=0.0045)
HMGCS2 | 204607_at
Expression level (log2)
Age
Smoker
non−smoker

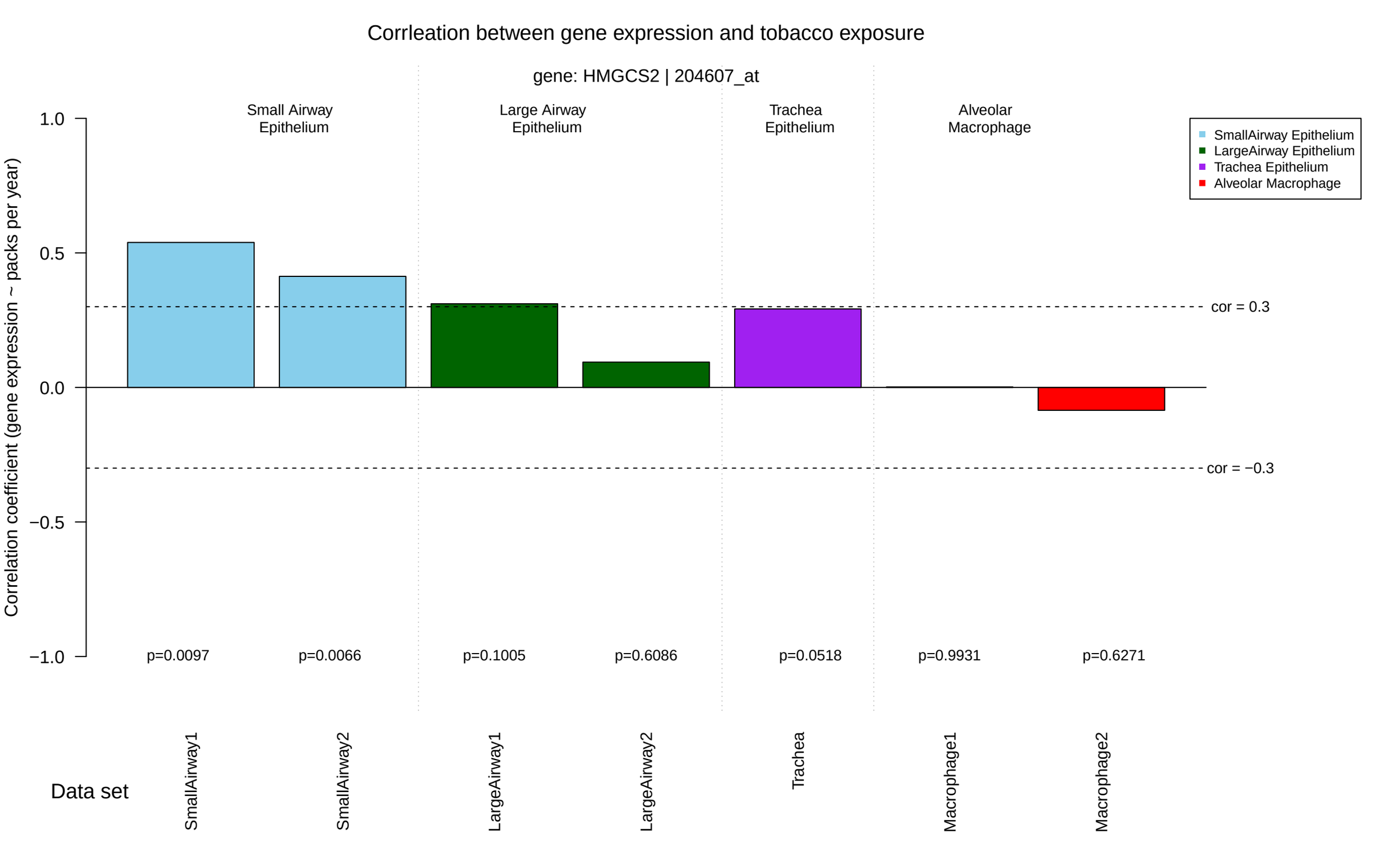

set: LargeAirway1 (p=0.3638)
HMGCS2 | 204607_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4053)
HMGCS2 | 204607_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1425)
HMGCS2 | 204607_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7999)
HMGCS2 | 204607_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9292)
HMGCS2 | 204607_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HMGCS2 | 204607_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0097
p=0.0066
p=0.1005
p=0.6086
p=0.0518
p=0.9931
p=0.6271
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2